

SIH26151 — Dark Web Threat Actor De-anonymization

FINALIZED Action Plan (Team's Selected Problem Statement)

Organization: National Technical Research Organisation (NTRO) **Category:** Software | **Theme:** Blockchain & Cybersecurity **Status:** FINAL PICK — this is the PS the team is building for SIH 2026

1. Official Problem Statement

Background: The dark web has become a preferred operating space for threat actors in the modern age, mainly because it lets them hide their identity behind Tor hidden services, which makes attribution of threat actors operating on darkweb the main challenge for any investigation. Such threat actors carry out a wide range of unlawful activities such as drugs and arms sale, stolen data and hacking services, money laundering, terror financing, etc. The objective of this problem statement is to build a system for the deanonymization of dark web threat actors and link them to suspect real-world entities.

Description: The system shall deanonymize dark web threat actors by continuously gathering their footprints from a range of sources (marketplaces, forums, deep web etc.) and linking them to the identifying information available on those sources. The system envisages three core capabilities. First, finding misconfigurations in Tor hidden services — such as exposed server-status pages, SSL certificates tied to clearnet domains, default service banners, descriptor inconsistencies, etc — and matching them with clearnet infrastructure to point to the likely origin servers. Second, mapping threat actors across multiple marketplaces into a single relationship graph of handles, PGP keys, wallets and trust links. Third, using AI-based analysis, including stylometric persona identification and behavioural profiling, to link rebranded or migrated personas to known threat actors. The system shall provide an analytical front end to query the database across a chosen timeline and shall work in an autonomous mode, drawing on available sources of good quality and reliability.

Expected Solution: An end-to-end system for the collection, storage, contextualization and querying (through GUI/dashboards) of dark web threat actor intelligence — covering actor profiles, identifiers (handles, PGP keys, wallets etc.), hidden service infrastructure indicators, persona linkages, attribution confidence, category, last scan date and source. The system shall also provide the facility to export the result set in CSV, JSON and report formats.

2. Solution Summary — What We're Building

An end-to-end pipeline that de-anonymizes dark web threat actors using **three independent signals fused into one confidence score:**

Infrastructure misconfiguration matching — cert/banner/descriptor leaks pointing to real-world servers

Timing-pattern correlation — response latency, jitter, and session-duration fingerprinting to catch co-located services even when infra signals don't match

Entity relationship graph — handles, PGP keys, wallets, trust links across marketplaces

Operational-style stylometry — text style *plus* behavioral habits (time zone, posting order, key rotation) that survive a deliberate rebrand

All four feed into an **adaptive confidence fusion layer** that re-weights signals when it detects the target is using countermeasures (e.g. trusts infra-matching less when timing looks deliberately padded).

This has been validated as a sound, well-differentiated approach — confirmed independently against academic literature (CARONTE, VeriDark-style stylometry, timing side-channel research) and judged to be ahead of what most competing teams will attempt, since most will implement only the baseline infra-matching + basic graph + generic text stylometry described literally in the PS, without the timing-signal and operational-habit layers.

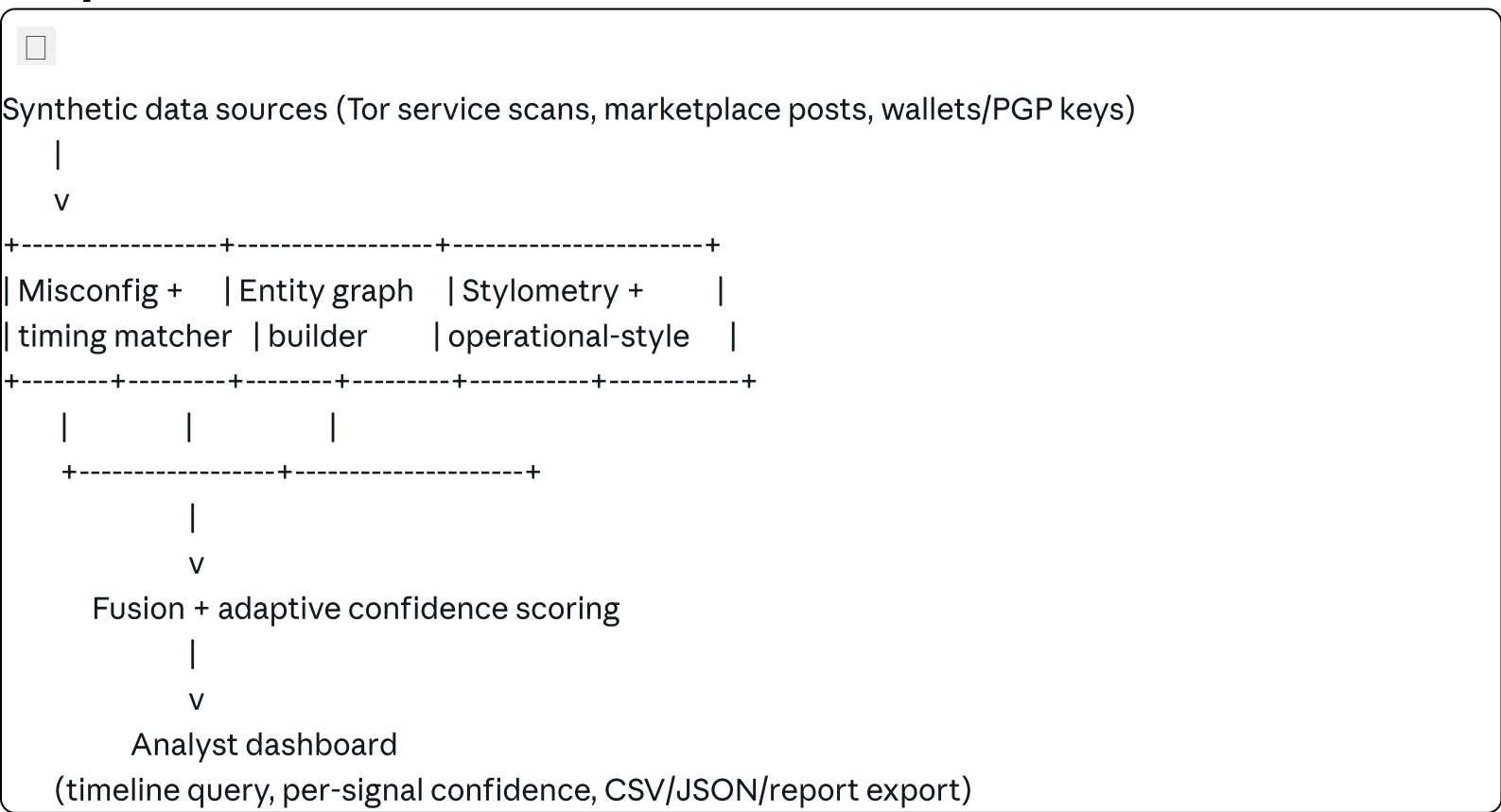
3. Core Differentiators (What Makes the System Different, Not the Team)

Stated purely as system behavior — this is what gets said in the pitch:

#	Differentiator	Why it matters
1	Timing-pattern correlation as a second infra signal	Catches co-located services even when certs/banners are clean
2	Adaptive confidence weighting	Down-weights unreliable signals when the system detects countermeasures (traffic shaping, padding) are in play
3	Operational-style stylometry	Persists through deliberate rebrands where pure text-style would fail
4	Honest "insufficient evidence" output	When signals conflict or are weak, the system says so explicitly rather than forcing a confident guess

Point 4 is a simplified, hackathon-feasible version of a more advanced idea (formal uncertainty theory) — implemented here as straightforward logic: if the fused confidence score falls in an ambiguous band, or if signals actively disagree, the output is explicitly flagged as "low confidence / conflicting evidence" rather than shown as a clean match.

4. System Architecture



5. Team Roles (6-member SIH team)

Role	Responsibility
Member 1	Misconfig + timing-pattern matcher
Member 2	Entity relationship graph builder
Member 3	Stylometry + operational-style classifier
Member 4	Fusion/confidence scoring + backend integration
Member 5	Dashboard (query UI, timeline filter, export)
Member 6	Synthetic dataset design, pitch deck, judge Q&A prep

6. Synthetic Dataset Design (Build This First)

Foundation of the whole demo — needs real, planted ground truth so results can be proven, not just claimed:

15-20 synthetic threat-actor personas, each with: a Tor hidden service, a marketplace handle, a PGP key, a wallet address

Split leak types deliberately:

Some personas: obvious infra leak (matching cert/banner) — proves baseline matching works

Some personas: *only* a timing-pattern leak, no cert/banner match — proves differentiator #1 is doing real work, not just decoration

3-4 rebranded personas: new handle, new marketplace appearance, but same underlying operational style (time zone, posting order, key rotation cadence) — proves differentiator #3

1-2 "no match" cases: a persona with genuinely weak/conflicting signals — proves differentiator #4 (the system should flag these as low-confidence, not force a guess)

This ground truth is what lets the team report real precision/recall numbers to judges instead of vague claims.

7. 36-Hour Execution Plan

Hours 0-8: Foundation

Hr 0-2: Team split confirmed; design the 15-20 synthetic personas and their leak types

Hr 2-4: Seed ground truth — rebranded personas + the "no match" ambiguous cases

Hr 4-6: Finalize entity graph schema + timing-feature schema + operational-style feature schema; repo and database setup

Hr 6-8: Confirm module ownership, environment set up for all 6 members

Hours 8-20: Core build

Hr 8-16: Misconfig matcher (cert/banner/descriptor matching) + timing-pattern correlation (jitter, latency distribution, session duration) as a second signal

Hr 8-14: Entity graph builder — handles, keys, wallets, trust-links into a relationship graph

Hr 8-16: Stylometry baseline (text-style features) + operational-style features (time zone, posting order, key rotation) — **highest-risk module, start early**

Hr 18-20: Mid-point integration sync — confirm all three modules produce outputs the fusion layer can consume in a shared schema

Hours 20-30: Fusion, dashboard, validation

Hr 20-26: Fusion and adaptive confidence scoring — combine the three signals, implement re-weighting logic, implement the "low confidence / conflicting evidence" honest-output path

Hr 22-28: Dashboard — timeline query filter, actor profile view, per-signal confidence breakdown, CSV/JSON/report export

Hr 28-30: Run the full pipeline against the synthetic dataset — confirm planted rebrand cases, timing-only cases, and ambiguous "no match" cases are all handled correctly; record real precision/recall numbers

Hours 30-36: Polish and pitch

Hr 30-33: Bug fixes and polish (buffer window — treat as such, don't plan new features here)

Hr 33-35: Pitch deck and demo script — explicitly state synthetic-data scope, explain the production pipeline path, prepare judge Q&A answers

Hr 35-36: Full dry-run of the demo with timing check

8. Risk Notes

Stylometry/operational-style module is the highest execution risk. Start at Hr 8, not later. If it's not producing usable signal by Hr 14-16, fall back to simpler features (word frequency, sentence length, posting-hour histogram) rather than chasing a more complex model. A working simple system beats a broken sophisticated one in a live demo.

State the synthetic-data scope proactively in the pitch. Don't let a judge discover the "this isn't live data" gap themselves — open with it, then explain the production path.

Don't attempt the advanced/research-grade upgrades (formal evidential-reasoning math, graph neural networks, transformer-based stylometry, hardware clock-skew fingerprinting) within the 36-hour window — these were surfaced during research as *future/production-grade* enhancements, not hackathon-feasible scope. Mention 1-2 of them only as a "future work" slide if time allows.

9. Anticipated Judge Questions

Question	Answer approach
"Will this work on real dark web data?"	Method is proven on synthetic data with known ground truth; production requires a live collection pipeline, which is a defined next phase, not a gap being hidden
"Is this legal?"	The system is a detection/analysis method, not a live scraping/access tool being demonstrated
"What's your accuracy?"	Report actual precision/recall against the planted synthetic ground-truth cases — real numbers, not estimates
"What if the system is wrong / signals disagree?"	Point to the honest low-confidence output path — the system flags conflicting evidence instead of forcing a guess
"How is this different from existing dark web intelligence tools?"	Existing commercial tools are mostly large indexed databases for manual analyst search; this system automates fusion of infra, timing, graph, and behavioral signals into one confidence score